# Beyond Cognitive Load:

## The Projection Principle at Planetary Scale
## Manufactured Trust, the Agent Byte, and the Economics of Admission

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# Abstract

This paper extends the keystone thesis [?] from the theory of a single faithful receipt to the physics of a planetary population of them. We do not restate the keystone's proofs; we cite them and build forward on three fronts. *First*, we make the Comprehension–Verification Gap quantitative, taking the autonomous build of $\sim 2.5 \times 10^6$ interior tokens as a concrete instance and separating what is *measured* (the interior token count, the receipt field count) from what is *estimated* (per-message hash wall-time), so the gap ratio $\sim 6 \times 10^5$ is stated with its provenance. *Second*, we formalize the industrial practice of *manufactured trust* — the simdjson/QLever treatment — as a general method: a correctness question that Rice's theorem [?] makes undecidable is retracted onto a decidable *oracle*. We give three oracles and their guarantees: differential agreement as mutual verification (with a false-accept bound), boundary fuzzing of the *total* admission retraction, and mutation-testing of the receipt chain, proving the *Mutant Kill Theorem* — under staged soundness and completeness, a mutant is killed *iff* the validator rejects it at the correct stage. *Third*, we develop the `agent8` projection: an MCP/A2A agent rendered as an eight-bit status byte. We prove that fleet admission is *branchless mask algebra* gating 64 agents per machine word per lane in a fixed seven-instruction reduction, show that a $10^{10}$-agent population is exactly 10 GB swept in one memory-bandwidth pass, and give the planetary Fermi estimate ($10^8$–$10^{10}$ admission decisions per second) establishing that only bit-parallel admission is affordable while comprehension-based admission is not. We close by formalizing the *antibody clause*: the overclaim set is empty as an invariant, and the admissible magnitude of any claim is bounded by the mechanism that receipts it — a bound this paper applies reflexively to its own claims. Every abstract object is anchored to running code in the `praxis` repository: the eight-lane `DenialPolarity` word, the eight-bucket `refusal` taxonomy, the staged `PowlReplayVerifier`, the BLAKE3 `OcelCausalFrame` chain, and the `agent8` byte.

# Contents

# Chapter 1

# Introduction: From the Gap to the Sweep

## 1.1 What the keystone settled, and what remains

The keystone thesis [**?**] settled the *qualitative* shape of trust among agents who cannot comprehend one another. Its results are three. The *Faithful Projection Theorem* [**?**, Thm. 3.2] establishes that a chained receipt committing law-bearing fields is faithful to lawful consequence over exactly those fields, under collision resistance. The *Comprehension–Verification Gap* [**?**, Thm. 4.2] establishes that boundary verification of a receipt costs $O(\kappa)$ — independent of interior dimension — while honest replay costs $O(n)$. The *Conservation of Consequence* [**?**, Thm. 3.5] establishes that every actuated artifact is the image of at least one admitted observation and occupies exactly one receipt-chain position. Together these say: a bounded verifier can trust an unbounded interior by reading a small faithful coordinate of it.

Those theorems are about *one* receipt and *one* verifier. A civilization is not one receipt. It is $10^{10}$ or $10^{12}$ agents, each emitting receipts continuously, each obliged to admit or refuse the others' outputs faster than any of them can be read. Three questions the keystone raised but did not exhaust are the subject of this paper.

(Q1) **How large is the gap, concretely?** The keystone proved the ratio diverges; it did not pin the constants. We take the 2.5-million-token build as an instance and compute the ratio with a clean line between measured and estimated quantities (Chapter **??**).

(Q2) **Where does the trust actually come from, mechanically?** The keystone named "differential agreement, adversarial fuzzing, invariants, and checked projections" as the engineering practice it was generalizing [**?**, Ch. 1], but treated them as motivation. We formalize them as a *general method* with provable guarantees (Chapter **??**).

(Q3) **What does admission cost at population scale?** The keystone's Proposition [**?**, Prop. 6.2] gave the agent byte and a one-line bandwidth argument. We derive the branchless admission algebra in full, prove its per-word gating, and give the planetary economics that make it the *only* affordable regime (Chapters **??** and **??**).

## 1.2 The one-line thesis of this paper

Everything below is a corollary of a single move, made three times. A question of the form *"is this interior correct / meaningful / trustworthy?"* is, by Rice's theorem [**?**], undecidable in general (this is the keystone's Theorem 2.1, on which we build). The industrial answer — and, we argue, the only scalable answer — is never to ask that question. It is to *retract* it onto a decidable surrogate: an equality between two oracles, a total classifier's labelled output, a hash comparison, a bitwise mask. Each surrogate is cheap, mechanical, and faithful over exactly what it commits. The keystone called this the projection principle for one receipt. This paper shows it is also the mutation oracle, the differential oracle, the fuzzing oracle, and — at population scale — a single AND of eight bit-planes sweeping the planet in one pass over memory.

*The gap is why comprehension fails; the sweep is what replaces it.*

**Structure.** Chapter **??** quantifies the gap on the instance. Chapter **??** formalizes manufactured trust and proves the Mutant Kill Theorem. Chapter **??** constructs the agent byte and proves branchless admission. Chapter **??** gives the planetary Fermi estimate. Chapter **??** formalizes the antibody clause. Chapter **??** concludes. Throughout, results proved are theorems; quantities that must be estimated are set as *Estimates* and labelled as such, following the keystone's discipline of claiming only what is true.

# Chapter 2

# The Comprehension–Verification Gap, Measured

The keystone proved (Theorem 4.2 there) that the ratio of comprehension cost to boundary-verification cost diverges as the interior grows. Divergence is a statement about a limit; a civilization runs on constants. This chapter supplies the constants for the canonical instance and, in doing so, fixes a methodological rule for the whole series: *every reported number is tagged measured or estimated.*

## 2.1 Measured and estimated quantities

**Definition 2.1** (Instance quantities)**.** For an executed manufacture $\sigma$ we distinguish:

- $T(\sigma)$, the *interior token count* — the number of machine-generated tokens in the full trace. This is *measured*: it is a count of logged records.
- $\kappa$, the *boundary field count* — the number of independently manipulable chunks in the receipt projection $\pi(\sigma)$ (Definition [**?**, Def. 3.1]). This is a *design constant*, fixed by the frame schema, not by the run.
- $t_{\mathsf{H}}$, the *per-frame recomputation wall-time* — the time to hash one frame and compare to the claimed chain value. This is *estimated* from published throughput of the hash primitive; it is not claimed as measured here.

*Remark.* The tag matters because the doctrine's credibility is exactly its refusal to launder estimates as measurements (the antibody clause, Chapter **??**). $T$ and $\kappa$ are hard numbers; $t_{\mathsf{H}}$ is a Fermi input. We never add a measured and an estimated number without saying so.

## 2.2 The instance

The canonical instance is the autonomous build of the keystone's Chapter 6: a fleet performed exploration, planning, and implementation whose interior consumed

$$T(\sigma) \;\approx\; 2.5 \times 10^6 \text{ tokens (measured)},$$

no meaningful fraction of which any agent or human read. Its receipt projection has the schema

$$\pi(\sigma) = \big(\underbrace{\text{verdict}}_{1\text{ bit}},\ \underbrace{h_+}_{256\text{ bits}},\ \underbrace{\varphi}_{1\text{ scalar}},\ \underbrace{r}_{1\text{ reason word}}\big), \qquad \kappa = 4 \text{ fields (design).}$$

The verdict is one bit; $h_+$ is the 256-bit BLAKE3 [?] chain head of the `OcelCausalFrame` sequence; $\varphi \in [0,1]$ is the `PowlReplayVerifier` fitness (Chapter **??**); $r$ is one of the eight refusal categories (`Identity`, `Capacity`, `Topology`, `Temporal`, `Lifecycle`, `Authorization`, `Prerequisites`, `Reserved`) or empty on accept.

**Theorem 2.1** (Instance gap)**.** *Let comprehension cost be* $\mathrm{Comp}(\sigma) = \Theta(T(\sigma))$ *and boundary-verification cost be* $\mathrm{Ver}_\partial(\sigma) = \Theta(\kappa)$*, as in [?, Thm. 4.2]. For the instance,*

$$\frac{\mathrm{Comp}(\sigma)}{\mathrm{Ver}_\partial(\sigma)} = \Theta\!\left(\frac{T(\sigma)}{\kappa}\right) = \Theta\!\left(\frac{2.5\times10^6}{4}\right) \approx 6.25\times10^5.$$

*Both inputs to the ratio are measured/design constants; the ratio is therefore a measured consequence, not an estimate.*

*Proof.* Immediate from [?, Thm. 4.2] with the two constants of Definition **??** substituted. Comprehension must ingest $\Theta(T)$ tokens; boundary verification reads the $\kappa = 4$ receipt fields and applies the fixed verifier $V$ of [?, Cor. 3.3]. The quotient is $T/\kappa$; substitution gives the figure. No estimated quantity enters. □

*Remark* (The gap is not a compression ratio)*.* Six hundred thousand is not the ratio of trace bytes to receipt bytes; it is the ratio of what a verifier would have to *comprehend* to the number of independent chunks it actually *reads*. The full trace can still be replayed at honest cost $O(n)$ ([?, Def. 4.1]); the gap concerns comprehension versus boundary inspection, and it grows without bound as $T$ grows while $\kappa$ stays fixed.

## 2.3 Boundary cost is invariant under interior growth

**Proposition 2.2** (Interior-independence of the boundary)**.** *Let* $\{\sigma_i\}$ *be manufactures sharing the frame schema of Definition* **??** *but with interior token counts* $T(\sigma_i) \to \infty$*. Then* $\mathrm{Ver}_\partial(\sigma_i) = \Theta(\kappa)$ *is constant in i, and the estimated per-message wall-time* $\kappa \cdot (t_{cmp}) + c \cdot t_{\mathsf{H}}$ *(for a constant number c of recomputed frames in a spot-check) does not depend on* $T(\sigma_i)$*.*

*Proof.* The projection $\pi$ has codomain of size $\le \kappa$ by [?, Def. 3.1]; the verifier $V$ reads only $\pi(\sigma_i)$. Neither the field count nor $V$'s input size is a function of $T$. A spot-check recomputes a fixed number $c$ of frames, each $O(1)$ ([?, Def. 4.1]), so its cost is $c \cdot t_{\mathsf{H}}$, again independent of $T$. □

**Estimate 2.1** (Per-message wall-time)**.** With BLAKE3 throughput on the order of 1–3 GB/s per core [?] and a frame of a few hundred bytes, $t_{\mathsf{H}} \sim 10^{-7}$ s per frame; a four-field boundary compare is a handful of machine words, $\ll 10^{-7}$ s. A per-message verification that inspects the boundary and recomputes $c \sim 10$ spot frames costs $\sim 10^{-6}$ s. These are order-of-magnitude figures from published primitive throughput, labelled as estimates; the point they support — constancy in $T$ — is the *proven* Proposition **??**, not the estimate.

# Chapter 3

# Manufactured Trust as a General Method

Engineers already trust software they cannot comprehend. A branchless vector parser classifies sixty-four input bytes per instruction [?]; an indexed triple store answers a query over $10^{11}$ facts by a join plan no one reconstructs [?]. Neither is trusted by reading it. Both are trusted because a battery of *decidable oracles* stands in for the undecidable question of correctness. This chapter turns that practice into theory. The unifying claim is the keystone's, sharpened: each oracle is an *admission retraction* — a total, terminating, decidable surrogate for a semantic question Rice's theorem forbids us to decide directly.

## 3.1 The decidable-oracle move

**Definition 3.1** (Correctness question and decidable oracle)**.** Let $f : D \to R$ be an implementation and $S \subseteq D \times R$ a specification. The *correctness question* is $Q_S(f) \equiv \forall x\,(x, f(x)) \in S$. A *decidable oracle* for $f$ on a finite test set $X \subseteq D$ is a total computable predicate $\Omega : X \to \{0, 1\}$ such that $\Omega(x) = 1$ is intended to witness $(x, f(x)) \in S$, and $\Omega$ terminates on every $x \in X$.

**Proposition 3.1** (Why an oracle is necessary, not a convenience)**.** *For $S$ a non-trivial semantic property, $Q_S(f)$ is undecidable. Hence no procedure decides $Q_S(f)$ for arbitrary $f$; any mechanical trust in $f$ must be trust in some $\Omega$ evaluated on a finite $X$, together with an argument bounding the residual risk on $D \setminus X$.*

*Proof.* $Q_S(f)$ quantifies a non-trivial semantic predicate over the language computed by $f$; by Rice's theorem [?] — the keystone's Theorem 2.1 — it is undecidable. A total $\Omega$ on finite $X$ always terminates and is therefore not a decision procedure for $Q_S$; it is a retraction of $Q_S$ onto the decidable fragment "$f$ passes $\Omega$ on $X$." Trust in $f$ factors through $\Omega$ and $X$. □

Proposition ?? is the whole chapter in one line: the three methods below are three constructions of $\Omega$, each with a different residual-risk argument.

## 3.2 Differential agreement as mutual verification

The simdjson/QLever pattern's first oracle is *differential*: run a second, independent implementation and compare outputs. Agreement is decidable; correctness is not.

**Definition 3.2** (Differential oracle)**.** Let $f, g : D \to R$ target the same $S$. The *differential oracle* is $\Omega_{f,g}(x) = [\, f(x) = g(x)\,]$. A test *passes* at $x$ iff $\Omega_{f,g}(x) = 1$.

A pass can be wrong in only one way: both implementations wrong *and* wrong identically. The next proposition bounds that event under an explicit independence assumption.

**Proposition 3.2** (False-accept bound for differential agreement)**.** *Fix $x \in D$. Suppose $f, g$ err at $x$ with probabilities $p_f, p_g$, that their errors are independent, and that conditioned on both erring their wrong outputs are drawn from a set of at least $M \geq 2$ distinguishable values with collision probability $\leq 1/M$. Then the probability that $x$ passes the differential oracle yet $(x, f(x)) \notin S$ satisfies*

$$\Pr[\,\Omega_{f,g}(x) = 1 \wedge (x, f(x)) \notin S\,] \;\leq\; \frac{p_f\, p_g}{M}.$$

*Over an independent test corpus $X$ of size $n$ sampling the same regime, the probability that some silent-agreement error survives all $n$ tests is at most $n\, p_f p_g / M$, and the probability that a* fixed *silently-wrong input is missed by a corpus that would have exercised it with per-input hit rate $q$ is $(1-q)^n \to 0$.*

*Proof.* A false accept at $x$ requires $f(x) \neq S$-correct value *and* $g(x) = f(x)$. The first has probability $p_f$. Given both err (probability $p_f p_g$ by independence), agreement requires the two wrong outputs to coincide, probability $\leq 1/M$. If $f$ is correct at $x$ but $g$ wrong, agreement fails and no false accept occurs; if $f$ wrong but $g$ correct, agreement fails likewise. Hence the joint event has probability $\leq p_f p_g / M$. The union bound over $n$ corpus points gives $n p_f p_g / M$. The miss bound is the probability a Bernoulli-$q$ trigger never fires in $n$ draws, $(1-q)^n$. □

*Remark* (Mutual verification is symmetric admission). $\Omega_{f,g}$ admits $f$'s output on the evidence of $g$ and simultaneously $g$'s on the evidence of $f$: neither comprehends the other's interior, exactly as two agents at fleet scale do not. The independence assumption is the load-bearing one — shared bugs (same author, same library, same wrong spec reading) violate it, which is why the practice pairs implementations of *maximally different provenance* (a scalar reference against a SIMD kernel; a naive triple scan against an indexed join). We state the assumption rather than hide it; the antibody clause (Chapter **??**) forbids claiming a bound whose hypothesis is unmet.

## 3.3 Fuzzing the admission boundary

The second oracle probes not $\mu$ but $\alpha$ itself: it searches for inputs where the admission retraction misbehaves. Here the projection principle pays a structural dividend. Because $\alpha$ is *total* — its totality is mechanized by wildcard-free `match` on a closed refusal enum (the foundations paper's Proposition on mechanized totality, realized in `refusal.rs` where `RefusalScenario::category` and `denial_lane` cover every variant with no catch-all arm) — every input has a defined, labelled outcome.

**Definition 3.3** (Admission-boundary fuzzing oracle)**.** Let $\alpha : \mathcal{O} \rightharpoonup \mathcal{O}^* \cup \{\perp\}$ be the total admission retraction. The *fuzzing oracle* $\Omega_\partial$ maps a generated input $o \in \mathcal{O}$ to 1 iff $\alpha(o)$ *terminates* and yields either an admitted $x \in \mathcal{O}^*$ with a well-formed artifact $\mu(x)$ or a refusal $\perp$ carrying a category in the eight-bucket taxonomy; $\Omega_\partial(o) = 0$ iff $\alpha$ crashes, diverges, or returns an unlabelled outcome.

**Proposition 3.3** (Fuzzing soundness for a total retraction)**.** *If $\alpha$ is total and terminating with codomain $\mathcal{O}^* \cup \{\perp\}$ and $\perp$ is refusal-with- category (refusal-as-data), then for every $o \in \mathcal{O}$, $\Omega_\partial(o) = 1$, and any observed $\Omega_\partial(o) = 0$ is a genuine defect in the retraction, not a property of the input.*

*Proof.* Totality gives termination on all $o$; the closed codomain gives, for each $o$, either $x \in \mathcal{O}^*$ (whence $\mu(x)$ is defined by boundedness, [?, Def. 3.1]) or $\perp$ with a category, since every refusal path maps through the wildcard-free classifier. Thus the disjunction defining $\Omega_\partial(o) = 1$ holds for every $o$, so $\Omega_\partial \equiv 1$ on a correct implementation. Contrapositively, an observed 0 — a crash, a hang, or an unlabelled return — contradicts totality or closedness and hence witnesses an implementation defect. □

*Remark* (Fuzzing tests the retraction, never the semantics)*.* Fuzzing cannot and does not test "did the agent understand the observation" — that is undecidable (Proposition **??**). It tests that the *retraction* is total, terminating, and labelling: exactly the properties the keystone requires of $\alpha$. This is why fuzzing an admission boundary is cheap and conclusive where fuzzing for "meaning" would be neither. The boundary between admitted and refused is a decidable surface; the fuzzer walks it.

## 3.4 Mutation-testing the receipt chain

The third oracle turns the tools on the verifier itself. *Mutation testing* [?, ?] injects a small fault (a *mutant*) and asks whether the test suite — here, the staged `PowlReplayVerifier` — detects it. A mutant that no test detects is *surviving*; one that is detected is *killed*. We formalize the verifier as a pipeline of stages and prove that killing is exactly rejection at the mutant's proper stage.

The `praxis` verifier runs four stages in order: *recompute* (rehash each frame), *linkage* (each frame's back-pointer equals the predecessor's chain head), *monotonic* (sequence indices strictly increase), and *POWL token replay* (the marking geometry of [?, Ch. 5], fitness $\varphi = 1$ iff no disabled firing). Model this abstractly.

**Definition 3.4** (Staged validator)**.** A *staged validator* is a pipeline $V = (V_1, \ldots, V_k)$ where each stage $V_i : \mathcal{S} \to \{\text{pass}\} \cup \{\text{reject}_i\}$ tests a decidable invariant $I_i \subseteq \mathcal{S}$: $V_i(s) = \text{pass} \iff s \in I_i$. The pipeline accepts $s$ iff every stage passes, i.e. $s \in \bigcap_i I_i$; on rejection it reports the least $i$ with $s \notin I_i$. Stage $V_i$ is *sound* if $V_i(s) = \text{reject}_i \Rightarrow s \notin I_i$ and *complete* if $s \notin I_i \Rightarrow V_i(s) = \text{reject}_i$.

**Definition 3.5** (Mutation operator and correct stage)**.** A *mutation operator* $m$ maps a valid subject $s^\star \in \bigcap_i I_i$ to a mutant $m(s^\star)$ that violates a non-empty set of invariants. The *correct stage* of $m$ is

$$\mathsf{s}(m) \;=\; \min\{\, i : m(s^\star) \notin I_i \,\},$$

the first invariant the mutant breaks. A mutant is *killed* by $V$, written $\text{Kill}(m) = 1$, iff $V$ rejects $m(s^\star)$.

**Theorem 3.4** (Mutant Kill Theorem)**.** *Let $V = (V_1, \ldots, V_k)$ be a staged validator in which every stage is sound and complete (Definition **??**), and let $m$ be a mutation operator with correct stage $\mathsf{s}(m)$ (Definition **??**). Then*

$$\text{Kill}(m) = 1 \quad \iff \quad V_{\mathsf{s}(m)} \textit{ rejects } m(s^\star),$$

*and moreover the least rejecting stage reported by $V$ equals* $\mathsf{s}(m)$. *Equivalently: a mutant is killed if and only if the validator rejects it at the correct stage.*

*Proof.* ($\Leftarrow$) If $V_{\mathsf{s}(m)}$ rejects $m(s^\star)$ then $V$ rejects (some stage rejected), so $\mathrm{Kill}(m) = 1$.

($\Rightarrow$) Suppose $\mathrm{Kill}(m) = 1$, so some stage rejects $m(s^\star)$; let $j$ be the least such. By soundness of $V_j$, $m(s^\star) \notin I_j$. By completeness of every stage $V_1, \ldots, V_{j-1}$, none of which rejected, $m(s^\star) \in I_i$ for all $i < j$. Hence $j$ is the *least* index with $m(s^\star) \notin I_i$, which is by definition $\mathsf{s}(m)$. Thus $j = \mathsf{s}(m)$ and the rejecting stage is $V_{\mathsf{s}(m)}$.

For the "moreover": $V$ reports the least rejecting index, which the argument identifies as $\mathsf{s}(m)$. Finally, killing occurs at all iff $\mathsf{s}(m)$ exists, i.e. iff $m$ violates some invariant; a mutation that violates no $I_i$ leaves $m(s^\star) \in \bigcap_i I_i$, is accepted, and is (correctly) not killed — it is an *equivalent mutant*, semantically identical modulo the committed invariants. □

**Corollary 3.5** (Diagnostic completeness of the chain)**.** *Under Theorem* **??**, *mutation testing of a sound-and-complete staged validator is not merely a pass/fail gate: the* index *of the killing stage certifies* which *law-bearing invariant the mutation broke. A mutant that flips a chain byte is killed at* recompute*; one that rewires a back-pointer at* linkage*; one that reorders frames at* monotonic*; one that forces a disabled transition at* POWL replay*. Surviving mutants are exactly the equivalent mutants (no committed invariant broken), so a non-empty surviving set that is* not *equivalent is a proof that the frame under-commits — the Faithful Projection converse of [***?***, Thm. 3.2].*

*Remark* (The three oracles are one principle)*.* Differential agreement (Def. **??**), boundary fuzzing (Def. **??**), and mutation kill (Def. **??**) are three instances of the decidable-oracle move (Def. **??**): each replaces an undecidable "is it correct" with a decidable check — an equality, a totality witness, a staged rejection. This is the projection principle of [**?**] applied to the *trust process* rather than the *trust artifact*. The receipt projects the interior; these oracles project the correctness question.

### 3.4.1 Autonomic Reconciliation and Visual Gap Measurement

The reconciler corrects structural drift in runtime environments by implementing the $A = \mu(O)$ loop.

**Definition 3.6** (Residual Vector and Dominant Dimension)**.** Let $O$ be environmental observations and $A$ the target artifact. The measurement $\mu(O)$ returns a residual vector $R \in \mathbb{R}^k$ with:

$$r_i \;=\; \mathrm{measured}_i \;-\; \mathrm{midpoint}(\mathrm{target}_i). \tag{3.1}$$

Reconciliation selects the dominant dimension:

$$\mathrm{dominant} \;=\; \arg\max_{i=1}^{k} |r_i|,$$

and applies a repair operator subject to `RepairBand` limits.

### 3.4.2 Git Worktrees and CI Gate Oracles

**Definition 3.7** (Isolated Retrofit Worktree)**.** An applier isolates modifications by spawning a decoupled working tree:

$$\texttt{git worktree add}\ \mathrm{temp\_path}\ \mathrm{phase\_branch}, \tag{3.2}$$

running validation gates and executing the auto-rollback protocol `git reset --hard baseline_sha` on failure to guarantee main branch cleanliness.

# Chapter 4

# The Agent Byte and Branchless Fleet Admission

We now scale from one manufacture to a population. The keystone's Construction 6.1 projected an agent onto eight bits; here we build that byte from the running `DenialPolarity` lanes, prove that fleet admission is branchless mask algebra gating 64 agents per word per lane, and compute the footprint and sweep time of a $10^{10}$-agent population exactly (footprint) and by estimate (time).

## 4.1 The agent8 byte from the denial lanes

The `praxis` refusal machinery exposes a denial word with eight lanes: the `ADMITTED` lane and seven denial lanes (`WATCHDOG_DRAINED`, `PRECONDITION_FAILED`, `SLA_BREACH`, `AUTHORIZATION_DENIED`, `RESOURCE_EXHAUSTED`, `OBJECT_LIFECYCLE_VIOLATION`, `CONFORMANCE_GATE_FAILED`), each mapping to a bucket of the eight-category taxonomy of `refusal.rs`. The agent byte is the per-agent projection of these lanes into a positive ("OK") polarity.

**Construction 4.1** (The agent8 byte)**.** Project each agent $a$ to a status byte $\mathbf{b}(a) \in \{0,1\}^8$ whose lanes are the OK-polarity complements of the denial lanes:

$$\mathbf{b}(a) = \big(\mathsf{admitted},\ \mathsf{evidence},\ \mathsf{budget},\ \mathsf{authority},\ \mathsf{healthy},\ \mathsf{conformant},\ \mathsf{receipted},\ \mathsf{replayable}\big),$$

with lane value 1 meaning the corresponding obligation holds and 0 meaning its denial lane is set (Construction [**?**, Con. 2.1], the denial monoid). Write $\mathbf{1} = \texttt{0xFF}$ for the all-OK byte. An agent is *admissible* iff $\mathbf{b}(a) = \mathbf{1}$, equivalently iff its 8-lane denial restriction is $\mathbf{0}$.

*Remark* (The byte is computed, not asserted). As in the keystone, the $\mathsf{receipted}$ and $\mathsf{replayable}$ lanes are set only by chained receipts whose frames commit the law-bearing fields; the byte is therefore a faithful projection ([**?**, Thm. 3.2]) of the agent's receipted lifecycle, not a self-report. A peer with context $\kappa$ reads a counterpart's entire lawful status in one byte-load.

## 4.2 Two layouts and the branchless admission sweep

A fleet of $N$ agents is a matrix of bits: $N$ agents $\times$ 8 lanes. Two storage layouts matter.

**Definition 4.1** (AoS and SoA layouts)**.** The *array-of-structures* (AoS) layout stores the 8-bit byte of each agent contiguously; a 64-bit machine word holds 8 agents' bytes. The *structure-of-arrays* (SoA), or *bit-plane*, layout stores each lane as its own bitset: lane $\ell$ is a bit-array $P_\ell \in \{0,1\}^N$, and a 64-bit word of $P_\ell$ holds the lane-$\ell$ bit of 64 agents.

**Theorem 4.1** (Branchless admission gates 64 agents per word)**.** *Under the SoA layout of Definition* **??***, the admissibility of* 64 *agents is computed by the single branchless word expression*

$$\mathsf{adm} \;=\; P_1 \,\&\, P_2 \,\&\, P_3 \,\&\, P_4 \,\&\, P_5 \,\&\, P_6 \,\&\, P_7 \,\&\, P_8,$$

*where each $P_\ell$ here denotes the* 64*-agent word of plane $\ell$ and* & *is bitwise AND. Bit $j$ of* $\mathsf{adm}$ *is* 1 *iff agent $j$ (of the* 64*) has all eight lanes OK, i.e. iff $\mathbf{b}(a_j) = \mathbf{1}$. The computation uses exactly* 7 *AND instructions, contains no data- dependent branch, and processes* 64 *agents; its cost is* $7/64 \approx 0.11$ *instructions per agent, independent of the lane* values*.*

*Proof.* Agent $a_j$ is admissible iff $\bigwedge_{\ell=1}^{8} \mathbf{b}(a_j)_\ell = 1$ (Construction **??**). Bit $j$ of $P_\ell$ equals $\mathbf{b}(a_j)_\ell$ by the SoA definition. Bitwise AND acts coordinatewise, so bit $j$ of $P_1 \& \cdots \& P_8$ equals $\bigwedge_\ell \mathbf{b}(a_j)_\ell$, which is 1 iff $a_j$ is admissible. A left-fold of AND over 8 operands is 7 binary ANDs. AND is a straight-line instruction with no branch, so the sequence is branchless; it reads 8 words (64 agents' worth of each lane) and writes one, regardless of the bits' values. The per-agent instruction count is 7/64. □

**Corollary 4.2** (Reduction to a single denial word)**.** *Equivalently, in denial polarity, let $D_\ell$ be the* 64*-agent word of denial lane $\ell$ ($D_\ell = \neg P_\ell$). Then the* denied *set is $\bigvee_\ell D_\ell$ and the admitted set its complement, matching the join-semilattice of the denial monoid ([***?***, Prop. 2.4]): fleet admission is the word-parallel join of denial lanes followed by one complement. Adding a ninth lane costs one more AND (resp. OR) per* 64 *agents.*

*Remark* (SIMD widens the word). Theorem **??** is stated for a 64-bit scalar word. A 512-bit SIMD register (AVX-512) executes the same 7 ANDs over 512 agents per lane, dropping the per-agent cost to $7/512 \approx 0.014$ instructions/agent. The algebra is unchanged; only the word widens. This is the same mechanism by which the vector JSON parser classifies 64 bytes per instruction [**?**] — admission is byte-classification lifted to agents.

**Construction 4.2** (Borrow-Safe Zero-Lane Detector)**.** Let $w \in \{0,1\}^{64}$ be a denial word comprising eight 8-bit lanes. Define the lane masks $L_{\text{high}} =$ `0x8080_8080_8080_8080` and $L_{\text{low7}} =$ `0x7f7f_7f7f_7f7f_7f7f`. The zero-lane bitmask $z(w) \in \{0,1\}^{64}$ is computed branchlessly by:

$$z(w) \;=\; \neg\Big(\big((w \,\&\, L_{\text{low7}}) + L_{\text{low7}}\big) \;\vee\; w\Big) \,\&\, L_{\text{high}}, \tag{4.1}$$

where $+$ is wrapping addition and $\vee, \&, \neg$ are bitwise operations. Bit 7 of lane $j$ in $z(w)$ is 1 iff lane $j$ in $w$ is exactly `0x00`, and 0 otherwise.

**Theorem 4.3** (SWAR Fleet Sweep Verification)**.** *Let $w$ be a* 64*-bit word packing* 8 *agents. If $w$ is gated against the broadcast required mask, the number of admitted agents is obtained by the popcount:*

$$\textit{admitted}(w) \;=\; \texttt{popcount}\big(z(w)\big).$$

*This sweeps* 8 *agents in one CPU instruction with zero branches and no lane borrow leakage.*

## 4.3 Footprint and sweep time of a population

**Proposition 4.4** (Population footprint is exact). *A population of $N = 10^{10}$ agents, each an 8-bit byte (Construction **??**), occupies*

$$N \times 8 \textit{ bits} \;=\; N \textit{ bytes} \;=\; 10^{10} \textit{ bytes} \;=\; 10 \text{ GB},$$

*in either layout (AoS packs 8 agents per 64-bit word; SoA packs the same 8 planes of $N$ bits). This is an exact arithmetic identity, not an estimate.*

*Proof.* Eight bits is one byte; $N$ agents is $N$ bytes; $10^{10}$ bytes = 10 GB under the decimal-GB convention. Both layouts store the same $8N$ bits, hence the same $N$ bytes. No estimated quantity appears. □

**Estimate 4.1** (One-pass sweep time is memory-bandwidth-bound). A full admission sweep reads all 8 planes once (10 GB total) and, by Theorem **??**, performs $7N/64 \approx 1.1 \times 10^9$ AND instructions for $N = 10^{10}$. At scalar issue rates ($\sim 10^{10}$ simple ops/s/core) the compute is $\sim 0.1$ s single-core and far less across cores and SIMD; at streaming memory bandwidth $B \sim 10^2$–$4 \times 10^2$ GB/s the read of 10 GB takes

$$t_{\text{sweep}} \;\approx\; \frac{10\,\text{GB}}{B} \;\approx\; 25\text{–}100\,\text{ms}.$$

Compute is dominated by memory traffic: the sweep is *bandwidth-bound*, not compute-bound. The instruction count is exact (Theorem **??**); the wall-time is an order-of-magnitude estimate from commodity bandwidth, labelled as such.

**Corollary 4.5** (Admission throughput). *From Estimate **??**, a single commodity server admits $N = 10^{10}$ agents in $\sim$25–100 ms, i.e. at $\sim 10^{11}$–$4 \times 10^{11}$ agent-admission decisions per second per server, bandwidth-bound. This is the supply figure the planetary economics of Chapter **??** weigh against the comprehension-based alternative.*

# Chapter 5

# Planetary Capacity and the Economics of Admission

We now weigh demand against supply for the two admission regimes — *bit-parallel* (the sweep) and *comprehension-based* (an LLM or human reading each observation) — and show the gap is not a matter of optimization but of feasibility. All figures in this chapter are Fermi estimates and are labelled; the *ordering* they establish is robust to an order of magnitude in any input.

## 5.1 Demand

**Estimate 5.1** (Planetary admission demand)**.** Take a population of $N \in [10^9, 10^{12}]$ agents (the keystone's fleet), each re-admitted — on receiving a peer message, on a lifecycle transition, on a watchdog tick — at a rate $\rho \in [1, 10^2]\,\mathrm{s}^{-1}$. The aggregate admission demand is

$$\Lambda_{\text{demand}} \;=\; N\rho \;\in\; \left[10^9,\, 10^{14}\right] \text{ decisions/s},$$

with a central estimate of $\sim 10^{11}$–$10^{12}$ decisions/s for $N \sim 10^{10}$, $\rho \sim 10$.

## 5.2 Supply, two regimes

**Estimate 5.2** (Comprehension-based supply ceiling)**.** A comprehension-based admission decision — reading the observation with an LLM and judging its meaning — costs on the order of $10^{-2}$–$10^0$ s of accelerator time per decision (the keystone's $\sim 10^{-2}$ s figure, [**?**, Prop. 6.2]), i.e. a per- accelerator rate of $\sim 1$–$10^2$ decisions/s. Even a planetary datacenter fleet of $\sim 10^6$–$10^7$ accelerators yields a comprehension-supply ceiling

$$\Lambda_{\text{comp}} \;\lesssim\; 10^7\text{–}10^9 \text{ decisions/s},$$

with a conservative central estimate of $\sim 10^8$ decisions/s.

**Estimate 5.3** (Bit-parallel supply)**.** By Corollary **??**, one commodity server supplies $\sim 10^{11}$ decisions/s bandwidth-bound. A modest rack of $\sim 10^2$ servers supplies $\sim 10^{13}$ decisions/s; the primitive is $\sim 10^7\times$ cheaper per decision than an LLM judgment ([**?**, Prop. 6.2]), so bit-parallel supply scales into the demand band of Estimate **??** on hardware of a single facility.

**Theorem 5.1** (Affordability separation)**.** *Let $\Lambda_{demand}$, $\Lambda_{comp}$, $\Lambda_{bit}$ be the demand and the two supplies. Under the estimates above,*

$$\Lambda_{comp} \;\sim\; 10^8 \;\ll\; \Lambda_{demand} \;\sim\; 10^{11}\text{–}10^{12} \;\lesssim\; \Lambda_{bit} \;\sim\; 10^{11}\text{–}10^{13}.$$

*Hence comprehension-based admission is under-provisioned by three to four orders of magnitude for a $10^{10}$-agent fleet, while bit-parallel admission meets demand on a single facility. The separation persists under an order-of-magnitude perturbation of any single input.*

*Estimate.* Substitute the central estimates of Estimates **??**–**??**. The comprehension ceiling $\Lambda_{\mathrm{comp}} \sim 10^8$ falls short of demand $\sim 10^{11}$–$10^{12}$ by $10^3$–$10^4$; the bit-parallel supply meets or exceeds it. The gap between the two supplies is the keystone's per-decision cost ratio $\sim 10^7$ scaled by population. A tenfold error in any one input (population, rate, per-decision cost, accelerator count) moves the inequality by at most one decade and does not reverse the three-to-four decade comprehension shortfall. □

**Corollary 5.2** (Only bit-parallel admission is affordable)**.** *A planetary control plane that re-admits its population continuously cannot be built on comprehension: no attainable accelerator fleet closes the demand gap. It can be built on the sweep of Chapter* **??***, whose cost is dominated by a single pass over* 10 GB *of memory. Comprehension is reserved, as in the keystone, for* sampled audit *at honest $O(n)$ replay cost — never for the per-message path.*

*Remark* (The true floor is bandwidth, and it is generous)*.* Because the sweep is bandwidth-bound (Estimate **??**), the ultimate limit on planetary admission is memory-system throughput, not logic. Bit-plane admission sits comfortably above the Landauer thermodynamic floor and orders of magnitude below the energy of a single comprehension decision; the physical budget of planetary admission is the cost of streaming a few tens of gigabytes, which any facility affords many times per second. Comprehension-based admission has no comparable floor because its cost scales with interior dimension per decision, exactly the quantity the projection principle exists to avoid paying.

# Chapter 6

# The Antibody Clause, Formalized

A thesis addressed to a trillion agents, making claims of planetary capacity, must carry its own guardrail against grandiosity — otherwise it is the very overclaim it warns against. The keystone stated this as the antibody clause [**?**, Prop. 6.4]. We formalize it as an invariant on claim magnitude and then apply it to this paper.

## 6.1 Claims, witnesses, and magnitude

**Definition 6.1** (Claim, receipt witness, magnitude)**.** A *claim* is a pair $c = (\phi_c, w_c)$ where $\phi_c$ is a proposition and $w_c \in \mathbb{R}_{\geq 0}$ its asserted *magnitude* (the size of what it asserts: a throughput, a population, a coverage fraction). A *receipt witness* for $c$ is a receipt $r(c)$ whose boundary projection is accepted by the fixed verifier, $V(\pi(r(c))) = 1$, and whose committed fields *entail* an attested magnitude $\widehat{w}_c$ — the largest value the receipt's law-bearing fields support under Faithful Projection ([**?**, Thm. 3.2]). A claim is *admissible* iff it has a receipt witness with $w_c \leq \widehat{w}_c$.

**Definition 6.2** (Overclaim set and the antibody invariant)**.** The *overclaim set* of a system emitting claims $\mathcal{C}$ is

$$\mathsf{Over} \;=\; \{\, c \in \mathcal{C} : \nexists\, r(c) \text{ with } V(\pi(r(c))) = 1 \text{ and } w_c \leq \widehat{w}_c \,\},$$

the claims with no receipt witness, or whose magnitude exceeds what their witness attests. The *antibody invariant* is $\mathsf{Over} = \varnothing$: this is the `docs-exceed-mechanism` defect class of the `praxis` definition-of-done, enforced as a gate rather than a guideline.

**Theorem 6.1** (Magnitude is bounded by the receipting mechanism)**.** *If a system maintains the antibody invariant* $\mathsf{Over} = \varnothing$*, then for every emitted claim $c$, its magnitude is bounded by its receipt witness:*

$$w_c \;\leq\; \widehat{w}_c \;=\; \max\{\, w : \textit{the committed fields of } r(c) \textit{ entail magnitude } w \,\},$$

*and $\widehat{w}_c$ is verifiable at boundary cost $O(\kappa)$ independent of the interior that produced $c$. Consequently no claim can exceed what its mechanism can receipt, regardless of its rhetoric.*

*Proof.* By $\mathsf{Over} = \varnothing$ (Definition **??**), every emitted $c$ has a receipt witness with $V(\pi(r(c))) = 1$ and $w_c \leq \widehat{w}_c$; the first conjunct is verifiable at $O(\kappa)$ by the Comprehension–Verification Gap

([?, Thm. 4.2], and Proposition ?? here), and $\widehat{w}_c$ is a function of the committed fields, which are faithful under collision resistance ([?, Thm. 3.2]). A claim with $w_c > \widehat{w}_c$ would lie in $\mathsf{Over}$, contradicting the invariant; hence $w_c \leq \widehat{w}_c$. The bound is on the magnitude a *receipt* attests, so it is independent of the unread interior. □

**Corollary 6.2** (Reflexive antibody: this paper's own claims)**.** *Each quantitative claim in this paper is either a theorem with a proof (a receipt in the logical sense: the derivation is its witness, verifiable by reading the proof, of bounded length) or an* Estimate *explicitly labelled and provisioned with its inputs (a receipt of the second kind: the magnitude is bounded by the stated Fermi inputs, and no larger value is asserted). The partition is exhaustive by construction: no numerical assertion appears unlabelled. Therefore this paper's overclaim set is empty, and by Theorem* **??** *the magnitude of each claim is bounded by its witness — a proof or a labelled estimate — as the invariant requires.*

*Proof.* Enumerate the paper's magnitude-bearing statements. The gap ratio $6.25 \times 10^5$ (Theorem **??**), the footprint 10 GB (Proposition **??**), the per-word gating 7/64 (Theorem **??**), and the magnitude bound itself (Theorem **??**) are theorems with proofs. The wall-time (Estimate **??**), sweep time (Estimate **??**), demand, supplies, and affordability separation (Estimates **??**–**??**) are labelled estimates carrying their inputs. Every magnitude falls in one class or the other; none is unlabelled; hence $\mathsf{Over} = \varnothing$ for this document. □

*Remark* (The guardrail is what licenses the scale)*.* This is the doctrine's discipline made self-applying: *a grand system without receipts is grandiosity; a grand system that receipts its own overclaims is machinery* [?, Ch. 6]. A paper claiming planetary capacity earns the claim only by receipting it — by proving what it can and labelling what it estimates. The antibody clause is not modesty; it is the mechanism that makes a large claim admissible in the sense of the Chatman equation.

# Chapter 7

# Fleet Application and Autonomic Reconciliation

## 7.1 The praxis-retrofit fleet applier

When a plan produces a code modification as its artifact, the artifact must be applied to a live fleet. The `praxis-retrofit` crate realizes this as a *receipted applier*: every application is gated, isolated in a Git worktree, and admitted through a CI oracle before any branch is advanced.

**Definition 7.1** (AST-gate isolation)**.** An *AST gate* is a syntactic predicate $g : \mathcal{T} \to \{0, 1\}$ on an abstract syntax tree $\mathcal{T}$ that is computable in $O(|\mathcal{T}|)$ time. The retrofit applier gates every proposed change through an ordered battery $(g_1, \ldots, g_m)$ of AST gates: the change is admitted iff all gates pass, i.e. $\bigwedge_{i=1}^{m} g_i(\mathcal{T}) = 1$. A failing gate returns a denial with the gate index $i$ and the offending AST node, constituting a *refusal with reason* in the sense of the denial monoid (Proposition **??**).

**Proposition 7.1** (AST-gate admission is total and terminating)**.** *For any finite AST $\mathcal{T}$ and finite gate battery $(g_1, \ldots, g_m)$ where each $g_i$ is a syntactic predicate, the battery evaluation terminates in time $O(m \cdot |\mathcal{T}|)$ and returns either* `Admitted` *or a refusal carrying the first failing gate index and the offending node. The retraction is total: it never diverges and never returns an unlabelled outcome.*

*Proof.* Each $g_i$ traverses the finite tree $\mathcal{T}$ once in $O(|\mathcal{T}|)$ time (no recursion below the finite tree can diverge on a finite input). The battery evaluates the gates in order; on the first failure it returns the structured denial; on all-pass it returns `Admitted`. The `RetrofitApplier` in `praxis-retrofit` wraps each gate result in the `DenialPolarity` framework, so the denial composes as a monoid element (Chapter **??**). □

**Definition 7.2** (Isolated worktree applier)**.** The retrofit applier isolates each application attempt in a fresh Git worktree:

`git worktree add` temp_path phase_branch,

applies the change, runs the CI oracle $(g_1, \ldots, g_m)$ (which includes `cargo build`, `cargo test`, `cargo clippy`), and on any gate failure executes the rollback protocol:

`git reset --hard` baseline_sha.

On all-pass, the worktree change is merged into the target branch, and the chain receipt for the application is minted with the worktree OID committed into the frame's `obj_refs` payload digest.

**Proposition 7.2** (Main branch cleanliness is an invariant). *Under Definition* **??***, the main branch is never advanced past a failing CI gate. The rollback protocol is unconditional on failure: it runs before the worktree is removed, restoring the branch to* `baseline_sha` *regardless of which gate failed. Hence the main branch satisfies all CI gates at every point in its history.*

*Proof.* The applier's critical section is:

1. create worktree,
2. apply change to worktree,
3. run gate battery,
4. on pass: merge; on fail: `git reset --hard baseline_sha` and abort.

Step 4's fail branch executes before the worktree is deleted, and `git reset --hard` on a worktree has no effect on the main working tree. The main branch pointer is advanced only in step 4's pass branch, which is never reached on a gate failure. Hence the main branch history is a subsequence of attempts all of which passed every gate. □

## 7.2 Visual gap measurement and the reconciler residual

**Definition 7.3** (Visual gap report). A *visual gap report* is the output of `praxis-reconciler`'s `measure_gap` function: for each reconcilable dimension $i \in \{1, \ldots, k\}$, the residual $r_i = \text{measured}_i - \text{midpoint}(\text{target}_i)$ and the dominant dimension $i^* = \arg\max_i |r_i|$ together with a rendered diff block for the discrepancy. The report is bounded: it has $k$ residual values and one dominant index, of total size $O(k)$ independent of the interior that generated the measurements.

**Proposition 7.3** (The visual gap report is a bounded projection). *Definition* **??** *gives* $\text{gap} : \mathbb{R}^k \to \mathbb{R}^k \times [k]$*, a projection from the $k$-dimensional measurement space onto the residual vector and its argmax. This projection has the same cost structure as the receipt projection: $O(k)$ to compute, $O(k)$ to verify. The repair operator* `apply_repair` *then acts on the dominant dimension $i^*$ subject to* `RepairBand` *bounds, producing at most one corrective actuation — a bounded deterministic manufacture (law M2 of the foundations paper).*

*Proof.* The residual vector is a linear map (subtraction of midpoint from measurement); it costs $O(k)$ arithmetic operations. The argmax is a single pass over $k$ values. The dominant dimension is one index; the repair acts on that dimension only. By Definition, the `RepairBand` is a finite closed interval, so the repair is a bounded projection onto that interval. □

# Chapter 8

# Conclusion

The keystone proved that a bounded verifier can trust an unbounded interior through a faithful projection. This paper carried that principle to the scale at which it is not an optimization but a precondition of coherence, and made it quantitative and mechanical at each step.

We measured the gap: on the canonical 2.5-million-token instance it is $\sim 6 \times 10^5$, with measured inputs and a proved consequence (Theorem **??**), and boundary cost is provably invariant as the interior grows (Proposition **??**). We formalized manufactured trust as the decidable-oracle move (Proposition **??**) and gave it three faces — differential agreement with a false-accept bound (Proposition **??**), boundary fuzzing of a total retraction (Proposition **??**), and mutation-testing with the Mutant Kill Theorem (Theorem **??**) that ties a kill to rejection at the correct stage and turns the verifier's stage index into a diagnosis (Corollary **??**). We built the agent byte from the running denial lanes (Construction **??**) and proved fleet admission is branchless mask algebra gating 64 agents per word in seven instructions (Theorem **??**), with a $10^{10}$-agent population an exact 10 GB swept in one bandwidth-bound pass (Proposition **??**, Estimate **??**). We weighed the planetary economics and proved the affordability separation (Theorem **??**): comprehension-based admission is under-provisioned by three to four orders of magnitude, and only bit-parallel admission scales (Corollary **??**). Finally we formalized the antibody clause as the invariant $\mathsf{Over} = \varnothing$ and proved that claim magnitude is bounded by the receipting mechanism (Theorem **??**), applying the bound reflexively to this paper (Corollary **??**).

The picture that results is severe and simple. At planetary scale, no one reads anyone. Trust is not comprehension; it is a hash comparison, an oracle equality, a total classifier's label, and — for the population as a whole — a single AND of eight bit-planes streamed across memory. The interior may be a trillion tokens or a trillion agents; the admission is a sweep, and the sweep is affordable. What makes the civilization coherent is not that its members understand one another. It is that each carries a faithful byte, and the byte can be checked in one pass.

*Beyond cognitive load; within admissible projection — at the scale of the sweep.*

# Appendix A

# Notation and Constants

## Symbols specific to this paper

| Symbol | Meaning |
|---|---|
| $T(\sigma)$ | interior token count of a manufacture (measured) |
| $\kappa$ | boundary field count / verifier context capacity (design constant) |
| $t_{\mathsf{H}}$ | per-frame recomputation wall-time (estimated) |
| $\Omega,\ \Omega_{f,g},\ \Omega_{\partial}$ | decidable oracle; differential; admission-boundary |
| $V = (V_1, \ldots, V_k)$ | staged validator; $I_i$ its stage invariants |
| $\mathsf{s}(m),\ \mathrm{Kill}(m)$ | correct stage of a mutation; kill indicator |
| $\mathbf{b}(a) \in \{0,1\}^8$ | agent status byte (Construction **??**) |
| $P_\ell,\ D_\ell$ | OK-polarity and denial-polarity bit-planes of lane $\ell$ |
| $B$ | streaming memory bandwidth |
| $\Lambda_{\mathrm{demand}}, \Lambda_{\mathrm{comp}}, \Lambda_{\mathrm{bit}}$ | admission demand; comprehension supply; bit-parallel supply |
| $\mathsf{Over},\ w_c,\ \widehat{w}_c$ | overclaim set; claimed and attested magnitude |

## Constants used, with provenance

| Quantity | Value | Provenance |
|---|---|---|
| Interior tokens $T(\sigma)$ | $\approx 2.5 \times 10^6$ | measured (keystone Ch. 6) |
| Boundary fields $\kappa$ | 4 | design (receipt schema) |
| Gap ratio $T/\kappa$ | $\approx 6.25 \times 10^5$ | proved (Thm. **??**) |
| Byte width | 8 bits | design (denial lanes) |
| Gating cost | 7/64 instr/agent | proved (Thm. **??**) |
| $10^{10}$-agent footprint | 10 GB | exact (Prop. **??**) |
| Sweep time | 25–100 ms | estimate (Est. **??**) |
| Bit-parallel supply | $\sim 10^{11}$/s/server | estimate (Cor. **??**) |
| Comprehension supply | $\sim 10^8$/s planetary | estimate (Est. **??**) |
| Per-decision cost ratio | $\sim 10^7$ | keystone (Prop. 6.2) |

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